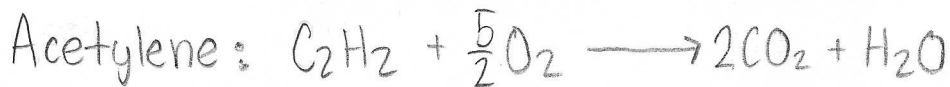


P6.47 Acetylene is used in welding torches, but it is much more expensive than methane. Compare the adiabatic flame temperature of acetylene to that of methane and see why acetylene is chosen instead of methane. (Assume stoichiometric amount of air is mixed with the fuel.)



	$\frac{J}{mol \cdot K}$
CO_2	37.0
H_2O	75.4 or 33.6
O_2	29.3
CH_4	35.7
C_2H_2	44.2

A: $\Delta \hat{H}_r^\circ = (-1)(228.2) + (-\frac{5}{2})(0) + (2)(-393.5) + (1)(-241.83) = -1257.03 \text{ kJ/mol}$

M: $\Delta \hat{H}_r^\circ = (-1)(-74.52) + (-2)(0) + (1)(-393.5) + (2)(-241.83) = -802.64 \text{ kJ/mol}$

$\Delta \hat{H}_r = n c_p \Delta T$ (pg 522) $\Delta T = \frac{\Delta \hat{H}_r}{n c_p}$

$\Delta H_f + \int_{T_1}^{T_2} c_p \Delta T = \Delta \hat{H}_r$

Do for all products w/ different T .

$\frac{5}{2} \left(\frac{0.79 \text{ mol } N_2}{0.21 \text{ mol } O_2} \right) = 9.40 \text{ mol } N_2$
 $2 \left(\frac{0.79 \text{ mol } N_2}{0.21 \text{ mol } O_2} \right) = 7.52 \text{ mol } N_2$

For reactants = to 0 b/c @ ref state
 so c_p is not needed for reactants.

M) $\Delta T = \frac{+802.64 \text{ kJ/mol}}{1(0.037) + 2(0.0336) + 7.52(0.0291) \frac{kJ}{mol \cdot K}} = 2484 \text{ K} = \Delta T$

A) $\Delta T = \frac{+1257.03 \text{ kJ/mol}}{2(0.037) + 1(0.0336) + 9.4(0.0291) \frac{kJ}{mol \cdot K}} = 3299 \text{ K} = \Delta T$

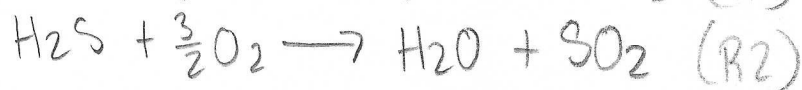
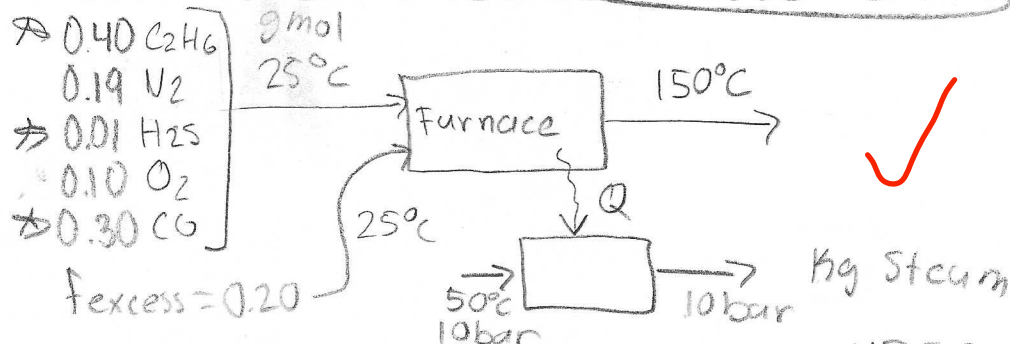
The change in temperature is greater for Acetylene therefore it's preferred over methane.

20/20

P6.48 A waste gas stream contains 40 mol% ethane (C_2H_6), 19 mol% N_2 , 1 mol% H_2S , 10 mol% O_2 , and 30 mol% CO . The gas is fed to a furnace along with 20% excess air and combusted. The waste stream entering the furnace is at $25^\circ C$. The air enters the furnace at $25^\circ C$. The flue gas exits the furnace stack at $150^\circ C$. Complete combustion can be assumed. The heat generated in the furnace is used to generate saturated steam at 10 bar, using boiler feedwater at $50^\circ C$ and 10 bar. Calculate the amount of steam generated in the furnace per gmol waste gas fed (kg steam per gmol gas).

BASIS

100 mol



O_2 NEEDED

$$\frac{7}{2}(40) + \frac{3}{2}(1) + \frac{1}{2}(30) = 156.5 \text{ mol}$$

$$156.5 - 10 = 146.5 \text{ of } O_2$$

$$1.20(146.5) = 175.8 \text{ of } O_2$$

$$175.8 \text{ of } O_2 \left(\frac{0.79 N_2}{0.21 O_2} \right) = 661.3 N_2$$

SPECIES BALANCE

$$C_2H_6 \quad 0 = 40 - \xi_1 \quad \xi_1 = 40$$

$$H_2S \quad 0 = 1 - \xi_3 \quad \xi_3 = 1$$

$$CO \quad 0 = 30 - \xi_2 \quad \xi_2 = 30$$

$$O_2 \quad n_{O_2} = 185.8 - \frac{7}{2}\xi_1 - \frac{1}{2}\xi_2 - \frac{3}{2}\xi_3 \quad n_{O_2, out} = 29.3$$

$$H_2O \quad n_{H_2O} = 0 + 3\xi_1 + \xi_3 \quad n_{H_2O} = 121$$

$$CO_2 \quad n_{CO_2, out} = 0 + 2\xi_1 + \xi_2 \quad n_{CO_2} = 110$$

$$SO_2 \quad n_{SO_2} = 0 + \xi_3 \quad n_{SO_2} = 1$$

$$N_2 \quad n_{N_2} = 19 + 661.3 \quad n_{N_2} = 680$$

$$Q = \Delta H$$

$$Q = \Delta H_f + \int_{T_1}^{T_2} C_p dT$$

w/respective Cp.

← NO SENSIBLE

$$\dot{Q} = \sum \dot{n}_e \hat{h}_e - \sum \dot{n}_i \hat{h}_i$$

$$\dot{Q} = 110(-393.5 + \int_{298}^{423} c_{p,CO_2} dT) + 121(-241.83 + I_{H_2O}) + 1(-296.81 + I_{SO_2})$$

$$+ 29.3(0 + I_{O_2}) + 680.3(0 + I_{N_2})$$

$$- [40(228.2) + 30(-110.53) + 1(-19.96) + 185(0)]$$

After Quickly doing integrals in graphing calculator by just changing cp value.

$$Q = -625 \text{ kJ/mol}$$

$$Q_{\text{to water}} = +625 \text{ kJ/mol}$$

$$\Delta H + \Delta E_k + \Delta E_p = Q + W$$

$$\Delta H = Q$$

$$Q = m(h_{\text{out}} - h_{\text{in}})$$

(Moron/Shapiro)

$$h_g = 2771 \text{ kJ/kg} \quad 10 \text{ bar}$$

$$h_f = 210 \text{ kJ/kg} \quad 50^\circ\text{C}$$

Phase Δ

$$\frac{Q}{\Delta h} = m = \frac{625}{2560} = 0.243 \text{ kg of Steam}$$

Produced per gmol

$$0.243 \frac{\text{kg}}{\text{gmol}}$$

20/20

0.85 Methane
0.15 Ethane

(3) A fuel gas containing 85 mol% methane and the balance ethane is burned completely with pure oxygen at 25°C, and the products are brought back down to 25°C.

(a) Suppose the reactor is continuous. Take a basis of 1 mol/s of fuel gas and combust with more than sufficient oxygen to calculate the heat rate transferred from the reactor per mole of fuel fed to the reactor

$\hat{Q} = \dot{Q}/\dot{n}$ in kJ/mol.

(b) Rework the problem for a constant-volume batch reactor. Use a basis of 1 mol of the fuel initially charged into the reactor and calculate the heat transfer per mole of fuel charged in kJ/mol.

(c) Briefly explain why the results in parts (a) and (b) do not depend on the percent excess oxygen.

$$\Delta \hat{H}_r^\circ = \sum_i \nu_i \hat{H}_{f,i}^\circ$$

$$\Delta \hat{H}_r^\circ = -\sum_i \nu_i \hat{H}_{c,i}^\circ$$

ISOTHERMAL

ISOCHORIC



$$\Delta \hat{H}_r^\circ = (-1)(-74.52) + (-2)(0) + (1)(-393.5) + 2(-285.84) = -890.66 \text{ kJ/mol}$$

$$\Delta \hat{H}_r^\circ = (-1)(-802.6) + (-2)(0) + (1)(0) + 2(+44) = -890.60 \text{ kJ/mol}$$

$$\Delta \hat{H}_r^\circ = (-1)(-83.82) + (-7/2)(0) + (2)(-393.5) + (3)(-285.84) = -1560.7 \text{ kJ/mol}$$

$$\Delta \hat{H}_r^\circ = (-1)(-1428.6) + (-7/2)(0) + (2)(0) + (3)(+44) = -1560.6 \text{ kJ/mol}$$

$$(A) \Delta \hat{H} + \Delta \hat{E}_k + \Delta \hat{E}_p = \hat{Q} + \hat{W} \quad \Delta \hat{H} = \hat{Q} \quad \sum \Delta \hat{H} = \hat{Q}$$

$$\hat{Q} = (0.85 \text{ mol Methane})(-890.6) + (0.15 \text{ mol Ethane})(-1560.7) = -991 \frac{\text{kJ}}{\text{mol}}$$

$$\hat{Q} = -991 \frac{\text{kJ}}{\text{mol}}$$

$$(B) \Delta \hat{H} = \Delta \hat{U} + P\Delta V \quad \text{or} \quad P\Delta V = nRT$$

$$\text{closed} \quad \Delta U + \Delta E_k + \Delta E_p = \hat{Q} + \hat{W}$$

$$\hat{Q} = \Delta U = \Delta H - nRT =$$

$$20/20$$

$$\text{CH}_4: \Delta U = -890.60 \frac{\text{kJ}}{\text{mol}} - (-2)(0.008314 \frac{\text{kJ}}{\text{mol} \cdot \text{K}})(298 \text{ K})$$

$$\Delta U = -885 \text{ kJ/mol}$$

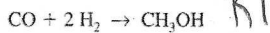
$$\text{C}_2\text{H}_6: \Delta U = -1560.6 - (-2.5)(0.008314 \frac{\text{kJ}}{\text{mol} \cdot \text{K}})(298 \text{ K})$$

$$\Delta U = -1554 \text{ kJ/mol}$$

$$\hat{Q} = 0.85(-885) + 0.15(-1554) = -986 \text{ kJ/mol}$$

(C) B/c the excess O₂ is not a part of heats of reaction. It doesn't react.

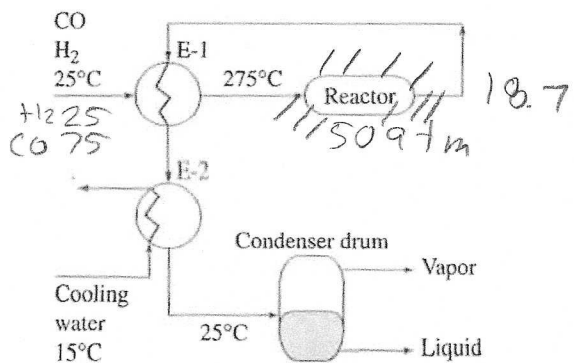
P6.68 Methanol is synthesized from CO and H₂ over a copper-based catalyst according to



A side reaction also occurs:



The reactants (75 gmol/s H₂, 25 gmol/s CO) enter the plant at 25°C and are heated to 275°C prior to being fed to the reactor. The reactor operates at 50 atm and is adiabatic. 80% of the CO fed to the reactor is converted to products. The production rate of CH₃OH out of the reactor is 18.7 gmol/s. The reactor products are cooled by heat exchange with the reactants entering the reactor. The stream is cooled further to 25°C with cooling water, then sent to a condenser drum, where liquid and vapor streams are separated and processed further. (Refer to sketch.) What is the temperature at the reactor outlet? What is the temperature of the product stream leaving the heat exchanger E-1?



$$f_{C,CO} = 0.80 \quad 0.80(25) = 20$$

$$\xi_1 + \xi_2 = 20$$

CO	$n_{CO} = 25 - \xi_1 - \xi_2$	$n_{CO} = 3$
H ₂	$n_{H_2} = 75 - 2\xi_1 - 3\xi_2$	$n_{H_2} = 33.7$
CH ₃ OH	$n_{CH_3OH} = 0 + \xi_1$	$\xi_1 = 18.7$
CH ₄	$n_{CH_4} = 0 + \xi_2$	$\xi_2 = 1.3$
H ₂ O	$n_{H_2O} = 0 + \xi_2$	$n_{CH_3OH} = 18.7$
		$n_{CH_4} = 1.3$
		$n_{H_2O} = 1.3$

$$\Delta \hat{H} = 0$$

$$C_{p,O} = 30.87 - 0.1285 T + 2.789 \times 10^{-5} T^2 - 1.272 \times 10^{-8} T^3$$

$$C_{p,H} = 27.14 + 0.0093 T - 1.381 \times 10^{-5} T^2 + 7.645 \times 10^{-9} T^3$$

$$C_{p,CH_3OH} = 21.14 + 0.07092 T + 2.587 \times 10^{-5} T^2 - 2.852 \times 10^{-8} T^3$$

$$C_{p,M} = 19.25 + 0.05213 T + 1.197 \times 10^{-5} T^2 - 3.596 \times 10^{-9} T^3$$

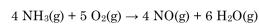
$$C_{p,W} = 32.24 - 0.001984 T + 1.055 \times 10^{-5} T^2 - 1.132 \times 10^{-8} T^3$$

no temp
calc.

-10

10/20

(5) In class we worked an example of burning ammonia and calculating the exit temperature of the gases.



The feed stream of 4 mol/s ammonia (NH₃) and 6 mol/s of oxygen (O₂) enters the reactor at 200°C and the product contain no ammonia and leaves at an unknown temperature T_{exit}(°C). Because of the limited range of applicability of the heat capacity relations we found two solutions for the exit temperature; namely, 2657 K and 3626 K. Rework the energy balance using better expression for the heat capacities. Go to:

<http://webbook.nist.gov/chemistry/>

and select Name under Search Options. Fill in the name of the compound and its phase and retrieve the data. You should use the Shomate equations for the ideal gas, constant-pressure heat capacity. Plot the Shomate equation for water from 25 °C to 6000 °C and compare with the expression given in Murphy on the graph. Show that the Shomate equation can be used down to 298 K and give reasonable results. What is the exit temperature when calculated using the Shomate equation? Explain why you now find only one solution.



Mole Balance

$$\dot{n}_{\text{NH}_3, \text{out}} = \dot{n}_{\text{NH}_3, \text{in}} - 4\dot{\xi} \quad \dot{\xi} = \frac{\dot{n}_{\text{NH}_3, \text{in}}}{4} = 1 \frac{\text{mol}}{\text{s}}$$

$$\dot{n}_{\text{O}_2, \text{out}} = \dot{n}_{\text{O}_2, \text{in}} - 5\dot{\xi} = 6 \frac{\text{mol}}{\text{s}} - 5(1 \frac{\text{mol}}{\text{s}}) = 1 \frac{\text{mol}}{\text{s}}$$

$$\dot{n}_{\text{NO}, \text{out}} = 4\dot{\xi} = 4 \text{ mol/s}$$

$$\dot{n}_{\text{H}_2\text{O}, \text{out}} = 6\dot{\xi} = 6 \text{ mol/s}$$

Energy Balance

INPUT/OUTPUT

SPECIES	$\dot{n}_{\text{in}} (\frac{\text{mol}}{\text{s}})$	$\hat{H}_{\text{in}} (\frac{\text{kJ}}{\text{mol}})$	$\dot{n}_{\text{out}} (\frac{\text{mol}}{\text{s}})$	$\hat{H}_{\text{out}} (\frac{\text{kJ}}{\text{mol}})$
NH ₃ (g)	4	$\hat{H}_1$	—	—
O ₂ (g)	6	$\hat{H}_2$	1	$\hat{H}_3$
NO(g)	—	—	4	$\hat{H}_4$
H ₂ O(g)	—	—	6	$\hat{H}_5$

Using Murphy B.17 (A, B, C, D)

$$\text{NH}_3: \hat{H}_1 = \int_{298\text{K}}^{473\text{K}} (C_p)_{\text{NH}_3(\text{g})} dT = 6.714 \frac{\text{kJ}}{\text{mol}}$$

$$\int_{298\text{K}}^{T_e(\text{K})} C_p dT \quad \text{at } 298\text{K}$$

$$\text{O}_2: \hat{H}_2 = \text{likewise}$$

$$\text{O}_2: \hat{H}_3 = \text{'' ''}$$

$$\text{NO}: \hat{H}_4 = \text{'' ''}$$

$$\text{H}_2\text{O}: \hat{H}_5 = \text{'' ''}$$

$$\left[\begin{array}{l} \frac{1}{1000} \left(A(T_e) + \frac{B}{2}(T_e)^2 + \frac{C}{3}(T_e)^3 + \frac{D}{4}(T_e)^4 \right) \\ \hat{H}(T_e) \end{array} \right] \leftarrow \text{likewise} \quad 8.514 \frac{\text{kJ}}{\text{mol}}$$

Find $\Delta \hat{H}_r^\circ$

	$\Delta \hat{H}_{f,i}^\circ \left(\frac{\text{kJ}}{\text{mol}} \right)$	$\Delta \hat{H}_{c,i}^\circ \left(\frac{\text{kJ}}{\text{mol}} \right)$
$\text{NH}_3(\text{g})$	-46.15	-316.8
O_2	0	0
$\text{NO}(\text{g})$	90.25	-90.2
H_2O	-241.83	0

B.3

Pg 647

$$\begin{aligned} \Delta \hat{H}_r^\circ &= \sum_i \nu_i \Delta \hat{H}_{f,i}^\circ = - \sum_i \nu_i \Delta \hat{H}_{c,i}^\circ \\ &= -905.38 \quad = -906.4 \end{aligned}$$

Energy Balance

$$\dot{Q} = 0 = \Delta \dot{H} = \sum_e \dot{n}_e \hat{H}_e(T_e) - \sum_i \dot{n}_i \hat{H}_i(T_i)$$

$$\begin{aligned} 0 &= \left(1 \frac{\text{mol}}{\text{s}}\right) \left(-905.38 \frac{\text{kJ}}{\text{mol}}\right) + \left[\left(1 \frac{\text{mol}}{\text{s}}\right) (\hat{H}_3(T_e)) + \left(1 \frac{\text{mol}}{\text{s}}\right) (\hat{H}_4(T_e)) + \left(6 \frac{\text{mol}}{\text{s}}\right) (\hat{H}_5(T_e)) \right] \\ &\quad - \left[\left(4 \frac{\text{mol}}{\text{s}}\right) (6.714 \frac{\text{kJ}}{\text{mol}}) + \left(6 \frac{\text{mol}}{\text{s}}\right) (5.31 \frac{\text{kJ}}{\text{mol}}) \right] \end{aligned}$$

incomplete - 10
10/20